

$$R(\theta) = \frac{v_0^2}{g} \sin(2\theta), \quad v_0 = 20 \text{ m s}^{-1}, \quad g = 9.81 \text{ m s}^{-2}$$

$$R(\theta) = 35 \text{ m} \quad \Rightarrow \quad f(\theta) = R(\theta) - 35 \text{ m} = 0$$